



DIOGO RIBEIRO

SOFTWARE ENGINEER

ABOUT ME:

I have been extremely interested in software since I was 14 years old. I developed a strong enthusiasm for the area of data management, storage systems and caching algorithms. In addition to my passion for software development I enjoy running and staying active in my free time.

CONTACTS

📞 966622079

✉️ ribeiroeribeiro2005@gmail.com

🌐 github.com/DIOGO4810

🌐 [linkedin.com/in/diogo-ribeiro-71742420b](https://www.linkedin.com/in/diogo-ribeiro-71742420b)

MY WEBSITE

🌐 <https://diogoribeiro.netlify.app/>

SKILLS

- Proficiency in programming technologies: C, Bash, MySQL, PostgreSQL, JavaScript, React, React Native, HTML, CSS, Python, Java, Git
- Strong understanding of OS concepts including memory hierarchy, process management, file systems, I/O and caching obtained by my academic background and practical experience in C/C++ data storage projects.

LANGUAGES

- Portuguese (Native Language)
- English: Proficient

PROFESSIONAL EXPERIENCE

• Researcher at Distributed Storage Research, HASLab/INESC TEC

Braga, Portugal — Jun 2025 – ongoing

- Contributed to an ongoing Modular I/O Library project at Haslab.
- Implemented caching mechanisms using CacheLib by developing C++ bindings exportable to C.
- Developed core components in C/C++, focusing on memory allocation, block alignment, and I/O performance.
- Designed and implemented AES-XTS-based encryption.
- Participated in performance evaluation and testing of the system.
- Collaborated on GitHub repository management and DevOps workflow.

• Event Host at Forum Braga

Braga, Portugal — Jun 2023 – Feb 2025

- Provided logistical and technical support, assisting teams and event organizers.
- Managed unforeseen situations and quickly resolved problems, ensuring the smooth running of events.
- Learned to work dynamically in a high-pressure environment, dealing with large audiences.

ACADEMIC EXPERIENCE

- Bachelor's degree in software Engineering at the University of Minho expected to graduate in 2026

PROJECTS

- Document indexing and search using only system calls in C (19.5/20): A client-server system for indexing and searching local documents, implemented solely using system calls, featuring caching and concurrency support.
- Backend simulation project of Spotify in C (19/20): A C project that involved CSV parsing and designing efficient data structures to store and manage information.
- Computer resource management in C (Self project): A C project that fetches hardware data from the OS to provide a simple and fast interface to manage computer resources.
- See this and other projects at github.com/DIOGO4810